

PhD Thesis Chapter Map — Block Diagram Edition

Self-Adaptive Model-Based Protection Systems (SAMBPS) for IBR-Penetrated Networks

Arun V. Eluvathingal | IIT Madras, EE Dept. | Version 3.0 · April 2026

Reading guide. **Diagram 1** — master one-page overview: SAMBPS layers → chapters → paper packages. **Diagram 2** — paper-flow detail: every package wired to its chapter, section, and journal. **Diagram 3** — Gantt + critical path (Q2 2026 → Q3 2028). **Tables** — TR allocation, contribution map, statistics.  done  draft/WIP  planned

Diagram 1 — Master Architecture Overview

SAMBPS Layers → Thesis Chapters → Paper Packages

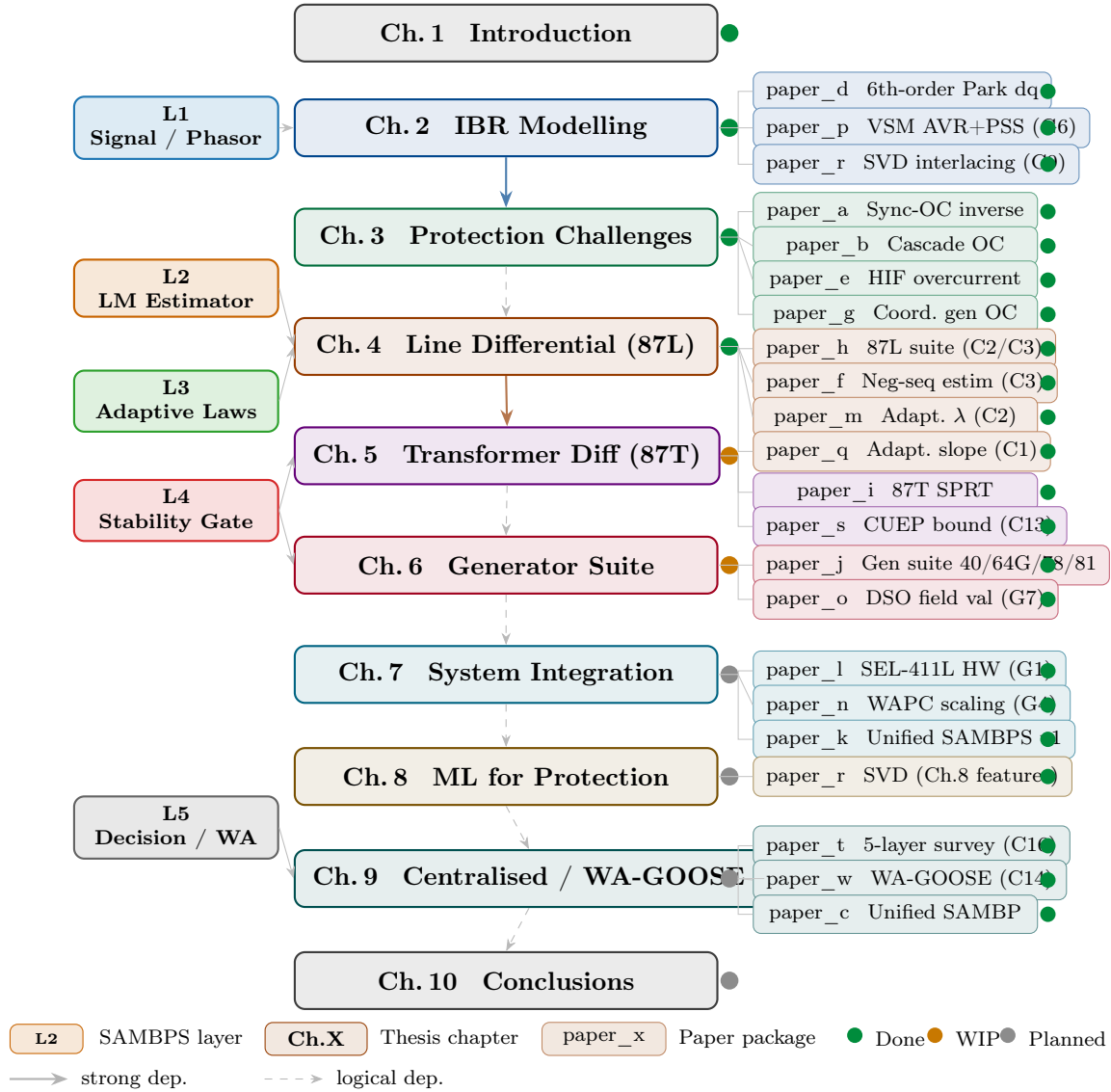


Diagram 2 — Paper-Flow Detail

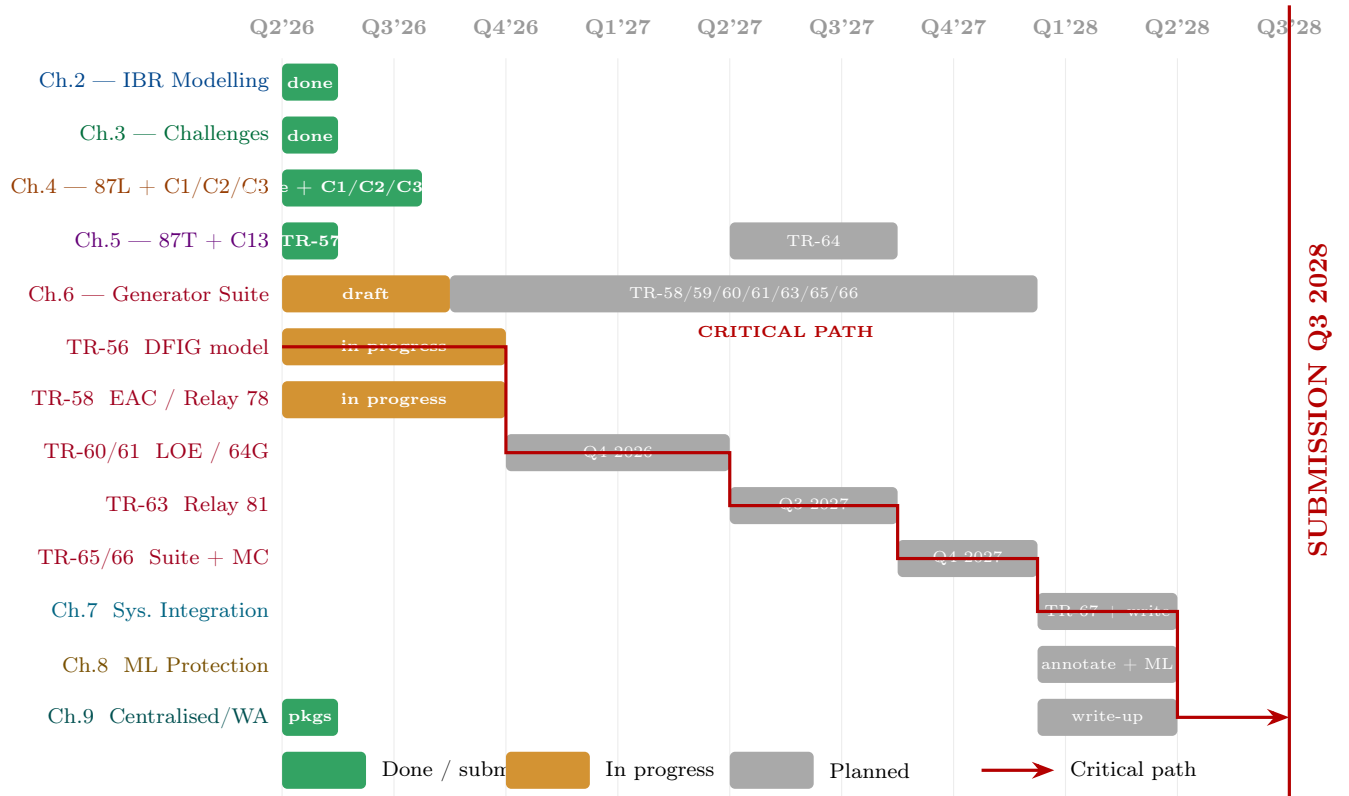
Every paper package: chapter → SAMBPS layer → primary layer → target journal

Package	Title (short)	SAMBPS Layer	Target journal
Ch.2 — IBR Modelling for Protection Analysis			
paper_d	6th-order Park dq: validity boundary $L^* < 150$ ms	PaperA L1	IEEE Trans. Power Syst.
paper_p	VSM 9-state: AVR+PSS extends L^* to 380 ms (G6)	G6 L1	IEEE Trans. Energy Convers.
paper_r	SVD interlacing: κ_n inheritance bound (C9)	C9 L2	SIAM J. Matrix Anal. Appl.
Ch.3 — Protection Challenges in IBR-Dominated Networks			
paper_a	Real-time inverse OC for SG feeders	PaperE L2	IEEE Trans. Power Del.
paper_b	Cascade OC coordination: Hilbert-Fortescue seq.	PaperB L2	IEEE Trans. Ind. Electron.
paper_e	HIF adaptive overcurrent on SG+IBR bus	PaperB L2	IEEE Trans. Power Del.
paper_g	Coordinated adaptive OC: parallel SGs	PaperG L2	IEEE Trans. Power Del.
Ch.4 — IBR-Adaptive Line Current Differential Protection (87L)			
paper_h	87L suite: adaptive threshold + 87LN + TDCS	PaperD L2	IEEE Trans. Power Del.
paper_f	Neg-seq estimator for asymmetric faults (C3)	C3 L2	IEEE Trans. Power Del.
paper_m	Adaptive forgetting factor $\lambda(t)$ (C2)	C2 L2	IEEE Trans. Power Del.
paper_q	Physics-constrained adaptive slope $k(t)$ (C1)	C1 L3	IEEE Trans. Autom. Control
Ch.5 — Transformer Differential Protection (87T)			
paper_i	3-hypothesis SPRT: inrush vs. IBR fault vs. CT-sat	TR-57 L2	IEEE Trans. Power Del.
paper_s	CUEP conservativeness + monotonicity (C13)	C13 L4	IEEE Trans. Power Syst.
Ch.6 — Generator Protection Suite (40 / 64G / 78 / 81 / 87G)			
paper_j	Generator suite: LOE + stator EF + QOS + freq + diff	PaperJ L2	IEEE Trans. Power Del.
paper_o	DSO field validation: 157 recordings, 3 failure modes (G7)	G7 L2	IEEE Trans. Power Syst.
Ch.7 — System Integration & Digital Twin Validation			
paper_l	SEL-411L relay hardware: 120 staged faults (G1)	G1 L2	IEEE Trans. Power Del.
paper_n	WAPC scaling: 3-substation ring network (G4)	G4 L5	IEEE Trans. Smart Grid
paper_k	Unified SAMBPS five-layer framework (C16 v1)	C16v1 L5	IEEE Trans. Power Syst.
Ch.8 — Physics-Informed ML for Fault Classification			
paper_r	SVD physics reduction as feature extractor for ML	C9 L2	SIAM J. Matrix Anal. Appl.
Ch.9 — Centralised Protection & WA-GOOSE Coordination			
paper_t	Unified 5-layer SAMBPS tutorial survey (C16)	C16 L5	Proc. IEEE
paper_w	WA-GOOSE quality-weighted consensus (C14)	C14 L5	IEEE Trans. Smart Grid
paper_c	Cross-element unified SAMBP framework	— L5	IEEE Trans. Power Syst.

Layer tag **L1** **L2** **L3** **L4** **L5** Signif. Adaptive Stability Decision/WA

Diagram 3 — Gantt Timeline & Critical Path

Q2 2026 → Q3 2028 submission. Red arrow = critical path.



Reference Tables

Table 1 — All 21 Paper Packages

Package	Title (short)	Ch.	SAMBPS	Target journal	pp
<code>paper_d</code>	6th-order Park dq validation	2	PaperA	IEEE Trans. Power Syst.	11
<code>paper_p</code>	VSM AVR+PSS: $L^*=380$ ms (G6)	2	G6	IEEE Trans. Energy Convers.	5
<code>paper_r</code>	SVD interlacing: κ_n (C9)	2,8	C9	SIAM J. Matrix Anal. Appl.	9
<code>paper_a</code>	Sync-OC inverse estimation	3	PaperE	IEEE Trans. Power Del.	8
<code>paper_b</code>	Cascade OC, Hilbert-Fortescue	3	PaperB	IEEE Trans. Ind. Electron.	6
<code>paper_e</code>	HIF adaptive overcurrent	3	PaperB	IEEE Trans. Power Del.	9
<code>paper_g</code>	Coordinated OC, parallel SGs	3	PaperG	IEEE Trans. Power Del.	7
<code>paper_h</code>	87L suite: harm.+neg-seq+TDCS	4	PaperD	IEEE Trans. Power Del.	10
<code>paper_f</code>	Neg-seq estimator 87L (C3)	4	C3	IEEE Trans. Power Del.	9
<code>paper_m</code>	Adaptive $\lambda(t)$ (C2)	4	C2	IEEE Trans. Power Del.	5
<code>paper_q</code>	Adaptive slope $k(t)$ (C1)	4	C1	IEEE Trans. Autom. Control	4
<code>paper_i</code>	87T SPRT inrush/IBR/CT-sat	5	TR-57	IEEE Trans. Power Del.	6
<code>paper_s</code>	CUEP conservativeness (C13)	5,6	C13	IEEE Trans. Power Syst.	5
<code>paper_j</code>	Gen. suite 40/64G/78/81/87G	6	PaperJ	IEEE Trans. Power Del.	6
<code>paper_o</code>	DSO field val. 157 rec. (G7)	6	G7	IEEE Trans. Power Syst.	5
<code>paper_l</code>	SEL-411L relay hardware (G1)	7	G1	IEEE Trans. Power Del.	4
<code>paper_n</code>	WAPC scaling val. (G4)	7	G4	IEEE Trans. Smart Grid	5
<code>paper_k</code>	Unified SAMBPS v1 (C16 v1)	7	C16v1	IEEE Trans. Power Syst.	5
<code>paper_t</code>	5-layer survey (C16)	9	C16	Proc. IEEE	6
<code>paper_w</code>	WA-GOOSE consensus (C14)	9	C14	IEEE Trans. Smart Grid	4
<code>paper_c</code>	Cross-element SAMBP framework	9	—	IEEE Trans. Power Syst.	7
Total (21 packages):					136

Table 2 — SAMBPS Contributions → Layer → Chapter → Paper

ID	Description	Layer	Ch.	Paper
C1	Adaptive slope: projected-gradient law, UUB Theorem	L3	4	paper_q
C2	Adaptive forgetting factor $\lambda(t)$	L2	4	paper_m
C3	Neg-seq estimator for 87L asymmetric faults	L2	4	paper_f
C9	SVD interlacing: κ_n inheritance bound	L2	2,8	paper_r
C13	CUEP conservativeness + monotonicity (IBR)	L4	5,6	paper_s
C14	WA-GOOSE quality-weighted consensus	L5	9	paper_w
C16	Unified 5-layer SAMBPS architecture	All	9	paper_t
G1	SEL-411L hardware validation	L2–L5	7	paper_l
G4	WAPC scaling validation	L5	7	paper_n
G6	VSM 9-state: $L^* = 380$ ms	L1	2,6	paper_p
G7	DSO field: 157 recordings, F1/F2/F3	L2–L5	6	paper_o

Table 3 — Statistics (April 2026)

Metric	Complete	Remaining
Paper packages (PDF + cover letter + contributions)	21 / 21	0
Total paper pages	136 pp	—
New TikZ figures (gap-paper sprint)	23	—
Journal papers submitted	7	+6–8 planned
Chapters: done / draft / planned	4 / 2 / 4	—
Technical reports complete	57	+10 (TR-56–67)
Monte Carlo trials to date	20k	+35k (gen. suite)
Critical path	TR-56 → 58 → 65 → 67 → Q3 2028	

Critical-path rule: No out-of-scope TR may begin before TR-56 reaches first-draft. Every new TR-68+ must declare: chapter home · section slot · failure mode · journal target — before development starts. Ch.7–9 write-up is gated on TR-65 (Q4 2027) and TR-67 (Q2 2028).